

## 6. Project Documentation

### 6.1 Your Approach to the Project

Our development process followed an agile, week-by-week structure with evolving responsibilities, using Google **docs** for planning and tracking tasks. We used **Godot** with **GDScript** for development, and **GitLab** for version control and collaboration through individual branches.

#### Development Timeline & Task Distribution:

- **Week of 24 April:**
  - We reviewed the Kanban board and began exploring key GDScript concepts such as card shuffling and pairing logic.
- **Week of 1 May:**
  - *Arda* created Godot project, set up the card scene, and added basic interaction through scripts.
  - *Esma* created the second card and implemented a basic win condition.  
We agreed on developing a prototype with 4 cards and started designing 12 card images.
- **Week of 8 May:**
  - *Arda* set up milestone tasks for all members.
  - *Esma* focused on card image designs.
  - *Nikunj and Rutvik* handled non-matching logic and resizable screen.
  - *Manan* started researching and designing a scoring system.
- **Week of 15 May:**
  - The prototype pitch extended the scope to include **touch input**.
  - *Manan* implemented timer and click tracking.
  - Work on difficulty levels and score calculation began.
- **Week of 17 May:**
  - We refined the point algorithm using base scores, click penalties, and time-based multipliers.
  - *Arda and Esma* implemented base mechanics for Level 0 (2 pairs).

- *Manan* led score logic with timers and click tracking.
- *Rutvik* began drafting the Requirements Documentation.
- **Week of 22 May:**
  - Requirements doc progressed (personas, user stories, glossary).
  - Coding focused on image resizing, level/menu logic, and APK research.
  - *Arda, Nikunj, and Manan* continued logic development.
- **Week of 28 May:**
  - Architectural documentation began (*Arda, Rutvik*).
  - Screen transitions and main screen setup by *Esma*.
  - Timer/click reuse logic was optimized by *Nikunj and Manan*.
- **Week of 5 June:**
  - Value passing between scenes/scripts was resolved (*Manan, Nikunj*).
  - Architectural doc continued with *Esma and Rutvik*.
- **Week of 12 June:**
  - Final UI scaling and mobile adjustments completed by *Esma and Arda*.
  - Documentation polishing began.
- **Week of 19 June:**
  - *Nikunj* added sound effects and background music.
  - *Rutvik and Manan* started preparing the final documentation and presentation.
  - UI polishing and scaling was finalized.
- **Week of 26 June:**
  - Documentation Sections 6 & 7 completed.
  - Final presentation and submission preparation.

## **6.2 Lessons Learned**

### **What Went Well:**

- Weekly task tracking using the Kanban board kept everyone aligned.
- Collaborative Git workflows with feature branches made development smoother.
- Godot's scene system made it easier to modularize cards, screens, and UI.
- The scoring and level system turned out to be both challenging and fun for players.

### **What Was Difficult:**

- Synchronizing work across branches occasionally led to merge conflicts.
- Designing a fair and scalable scoring system that works across all levels took several iterations.
- UI scaling across devices (especially for mobile) required additional effort.
- Late-stage adjustments (e.g., sound effects, screen transitions) compressed testing time.
- Need of adding Touch input increased the scope.

### **What We'd Do Differently Next Time:**

- Define screen layout and resolution targets earlier to reduce rework.
- Involve real test users (e.g., children or parents) sooner in the process.
- Automate testing for score logic and timer validation.
- Start documentation (especially architecture) earlier to avoid last-minute pressure.